

Understanding the Interquartile Range (IQR)

A Comprehensive Guide to Robust Statistical Dispersion

In statistics, **IQR** stands for **Interquartile Range**. It is a fundamental measure of statistical dispersion—a metric that quantifies how spread out the values in a dataset are. Mathematically, it is calculated as the difference between the 75th percentile (Third Quartile, or **Q3**) and the 25th percentile (First Quartile, or **Q1**).

Essentially, the IQR captures the range of the **middle 50%** of your data distribution, making it an exceptional tool for understanding core variability without visual distortion from extreme values.

Why Use IQR?

Unlike the total range (maximum value minus minimum value) or the standard deviation, the IQR is highly **robust against outliers**. Because it focuses strictly on the central half of the observations, a few unusually high or low numbers will not distort the metric. This property makes the IQR the foundational visual anchor for generating **Box Plots** and the mathematical benchmark for identifying anomalies.

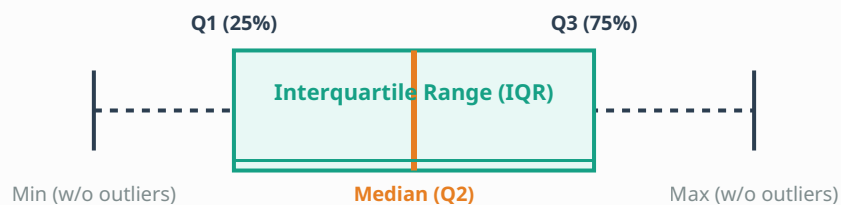


Figure 1: Anatomy of a Box Plot highlighting the IQR region.

How to Calculate IQR (Step-by-Step)

To compute the IQR manually or structurally within an algorithm, follow these four chronological steps:

- 1 Order the dataset:** Arrange all numerical observations sequentially from smallest to largest.
- 2 Locate the Median (Q2):** Identify the precise middle value, which bisects the entire dataset into an upper and lower half.
- 3 Isolate the Quartiles:**
 - **Q1 (25th percentile):** Find the median of the lower half of the dataset.

- **Q3 (75th percentile)**: Find the median of the upper half of the dataset.

4 Perform the Subtraction: Deduct the lower quartile from the upper quartile to yield the final range.

$$IQR = Q3 - Q1$$

A Concrete Practical Example

Consider an operational case study where a local coffee shop records the total number of daily espresso specialty drinks sold over a 9-day observation window:

7, 12, 15, 11, 4, 18, 14, 11, 30

Step 1: Sort the data sequentially

4, 7, 11, 11, 12, 14, 15, 18, 30

Step 2: Isolate the Median (Q2)

Because there are 9 distinct data points (an odd sample size), the exact middle number sits at the 5th position:

$$\text{Median (Q2)} = 12$$

- **Lower Dataset Split:** 4, 7, 11, 11
- **Upper Dataset Split:** 14, 15, 18, 30

Step 3: Determine Lower and Upper Quartiles (Q1 & Q3)

With an even count of 4 items in each split, we locate the respective sub-medians by averaging their center-most pairings:

- **Q1 Calculation:** Average the middle two values of the lower half (7 and 11):
$$(7 + 11) \div 2 = 9$$
- **Q3 Calculation:** Average the middle two values of the upper half (15 and 18):
$$(15 + 18) \div 2 = 16.5$$

Step 4: Compute Final IQR Value

$$IQR = 16.5 - 9 = 7.5$$

Statistical Interpretation: The central 50% of daily espresso sales spans a highly consistent range of 7.5 drinks. Notice that the exceptionally high volume day (30 drinks) had absolutely zero distortion effect on this core dispersion measurement.

Bonus: Algorithmic Outlier Detection

Statisticians and data scientists rely heavily on the standardized "**1.5 × IQR Rule**" to flag anomalies. An observation structurally qualifies as an outlier if it falls outside the following mathematical boundaries:

- **Lower Bound fence:** $Q1 - (1.5 \times IQR)$
- **Upper Bound fence:** $Q3 + (1.5 \times IQR)$

Applying this formula to our coffee shop data:

$$\text{Upper Boundary Fence} = 16.5 + (1.5 \times 7.5) = 16.5 + 11.25 = 27.75$$

Because the recorded maximum observation of **30** exceeds our established fence threshold of 27.75, it formally registers as a statistical outlier.